

Power Spectral Density

sample funcⁿ $x(t)$ of $x(t)$. Let $x_T(t)$ be defined as portion of $x(t)$ b/w $\lim -T$ & $+T$.

$$x_T(t) = \begin{cases} x(t) & \text{for } -T < t < T \\ 0 & \text{o/w} \end{cases}$$

• for finite value of $x_T(t) = 1 \quad \int_{-T}^T |x_T(t)| dt < \infty$

• $F[x_T(t)] = X_T(\omega) = \int_{-\infty}^{\infty} \int_{-T}^T x_T(t) e^{j\omega t} dt$

• $x_T(t) = x(t) \Rightarrow (-T, T)$

$$X_T(\omega) = \int_{-T}^T x(t) e^{-j\omega t} dt$$

$$E(T) = \int_{-T}^T x_T^2(t) dt = \int_{-T}^T x^2(t) dt$$

$$E(T) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X_T(\omega)|^2 d\omega$$

$$P_{xx}(T)_{\text{avg}} = \frac{1}{2\pi T} \int_{-T}^T x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|X_T(\omega)|^2}{2T} d\omega$$

$$\text{Avg Power} = \frac{1}{2AT} \int_{-T}^T x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|x_T(\omega)|^2}{2T} d\omega$$

$$P_{xx}(T) = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-\infty}^{\infty} E \left\{ \frac{|x_T(\omega)|^2}{2T} \right\} d\omega$$

$$P_{xx}(T) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\lim_{T \rightarrow \infty} \frac{E \{ |x_T(\omega)|^2 \}}{2T} \right] d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) d\omega$$

Power of random process

$S_{xx}(\omega)$ power density funcⁿ.

$$S_{xx}(\omega) = \lim_{T \rightarrow \infty} \frac{E \{ |x_T(\omega)|^2 \}}{2\pi}$$

Wiener Khinchin Relation

(Rel b/w PS & ACF).

For a given WSS $x(t)$, the ACF & $S_{xx}(\omega)$ form a fourier transform pair.

$$S_{xx}(\omega) = F[R_{xx}(z)] = \int_{-\infty}^{\infty} R_{xx}(z) \cdot e^{-j\omega z} dz$$

$$R_{xx}(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) e^{-j\omega t} d\omega$$

$$S_{xx}(\omega) = \lim_{T \rightarrow \infty} \frac{E \{ |x_T(\omega)|^2 \}}{2T}$$

$$x(t)x^*(t) = |x(t)|^2$$

$$S_{xx}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2T} E \{ x_T^*(\omega) x_T(\omega) \}$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} E \left\{ \int_{-T}^T x(t_1) dt_1 e^{j\omega t_1} \cdot \int_{-T}^T x(t_2) dt_2 e^{-j\omega t_2} \right\}$$

$$S_{xx}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \int_{-T}^T E [x(t_1)x(t_2)] e^{-j\omega(t_2-t_1)} dt_2 dt_1$$

$$S_{xx}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \int_{-T}^T R_{xx}(t_1, t_2) e^{-j\omega(t_2-t_1)} dt_2 dt_1$$

$$t_1 = t \quad dt_1 = dt \quad t_2 - t_1 = \tau \\ dt_2 = d\tau$$

$$t_2 + T = \tau \quad t_2 = \tau - T \\ \tau = T - t$$

$$t_1 - T = \tau \quad t_2 = \tau + T$$

$$S_{xx}(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T-t}^{T-t} \int_{-T}^T R_{xx}(t, t+\tau) e^{-j\omega\tau} d\tau dt$$

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{xx}(\tau) dt \left\{ e^{-j\omega\tau} d\tau \right\}$$

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$$S_{xx}(\omega) = \int_{-\infty}^{\infty} A[R_{xx}(t, t+\tau)] dt$$

$$A[R_{xx}(t, t+\tau)] = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_{xx}(t, t+\tau) dt$$

✓
Avg ACF

WSS

$$A[R_{xx}(t, t+\tau)] = R_{xx}(\tau)$$

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j\omega\tau} d\tau$$

Prop PSD

- ① 0 freq val of PSD of WSS
total area under ACF.

$$S_{xx}(0) = \int_{-\infty}^{\infty} R_{xx}(\tau) d\tau$$

- ② PSD of WSS is always non-neg.

$$S_{xx}(\omega) \geq 0$$

- ③ RMS of WSS = total area under graph of SD

$$E[x^2(t)] = R_{xx}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) d\omega$$

$$R_{xx}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) e^{-j\omega\tau} d\omega$$

$$\tau = 0 \quad R_{xx}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{xx}(\omega) d\omega = E$$

$$= E[x^2(t)]$$

④ PSD of ~~an~~ real valued R.P. is even func of freq -

$$S_{xx}(\omega) = S_{xx}(-\omega)$$

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j\omega\tau} d\tau$$

$$\omega = -\omega \quad S_{xx}(-\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{j\omega\tau} d\tau$$

$$\tau = -\tau \quad S_{xx}(-\omega) = \int_{-\infty}^{\infty} R_{xx}(-\tau) e^{-j\omega\tau} (-d\tau)$$

$$S_{xx}(-\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j\omega\tau} d\tau$$

$$S_{xx}(\omega) = S_{xx}(-\omega)$$

⑤ $S_{xx}(\omega)$ is always real $x_T(\omega)$ is real $x_T(\omega) = E[x_T(t)]$

(6) PSD of derivative $\left(\frac{dx(t)}{dt}\right)$ is ω^2 times P.S.D. of $x(t)$.

$$S_{xx'}(\omega) = \omega^2 S_{xx}(\omega)$$

Bandwidth

$$f_x(\omega) = \frac{S_{xx}(\omega)}{\int_{-\infty}^{\infty} S_{xx}(\omega) d\omega}$$

total area of $f_x(\omega) = 1$

The normalized PS behaves as PDF & measure of its spread is RMS bandwidth

$$\omega_{RMS}^2 = \frac{\int_{-\infty}^{\infty} \omega^2 S_{xx}(\omega) d\omega}{\int_{-\infty}^{\infty} S_{xx}(\omega) d\omega}$$

low pass
random
process

Band Pass RP

$$\omega_{RMS}^2 = \frac{4 \int_0^{\infty} (\omega - \omega_0)^2 S_{xx}(\omega) d\omega}{\int_0^{\infty} S_{xx}(\omega) d\omega}$$

RMS Bandwidth

$$\text{mean freq } \bar{\omega}_0 = \frac{\int_0^{\infty} \omega S_{xx}(\omega) d\omega}{\int_0^{\infty} S_{xx}(\omega) d\omega}$$

LTI

$$\text{WSS } \Rightarrow x(t) \rightarrow \boxed{\text{LTI}} \rightarrow y(t)$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$

$$E[y(t)] = E \left[\int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau \right]$$

$$E[y(t)] = \int_{-\infty}^{\infty} h(\tau) E[x(t-\tau)] d\tau$$

$$E[x(t-\tau)] = E[x(t)]$$

$$E[y(t)] = \bar{x} \int_{-\infty}^{\infty} h(\tau) d\tau$$

$$h(t) \xleftrightarrow{FT} H(\omega)$$

zero impulse
fun...

$$H(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau \Rightarrow H(0)$$

$$H(0) = \int_{-\infty}^{\infty} h(\tau) d\tau$$

$$E[y(t)] = \bar{x} H(0)$$

FT of $h(t)$ where
 $\omega = 0$

$$E[y^2(t)] = E \left[\int_{-\infty}^{\infty} h(\tau_1) x(t-\tau_1) d\tau_1 \int_{-\infty}^{\infty} h(\tau_2) x(t-\tau_2) d\tau_2 \right]$$

$$E[y^2(t)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E[x(t-\tau_1) x(t-\tau_2)] h(\tau_1) h(\tau_2) d\tau_1 d\tau_2$$

WSS $R_{xx}(t, t_2) = E[x(t) x(t_2)]$

$$E[y^2(t)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{xx}(\tau_1 - \tau_2) h(\tau_1) h(\tau_2) d\tau_1 d\tau_2$$

ACP of $Y(t)$

$$R_{YY}(t, t+\tau) = E \left[Y(t) Y(t+\tau) \right]$$

$$R_{YY}(t, t+\tau) = E \left[\int_{-\infty}^{\infty} h(u) X(t-u) du \int_{-\infty}^{\infty} h(z_2) X(t+\tau-z_2) dz_2 \right]$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E \left[X(t-u) X(t+\tau-z_2) \right] h(u) h(z_2) du dz_2$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{XX}(z + u - z_2) h(u) h(z_2) du dz_2$$

$$R_{YY}(\tau) = R_{XX}(\tau) * h(-\tau) * h(\tau)$$

CCF

$$R_{XY}(t, t+\tau) = E \left[X(t) Y(t+\tau) \right]$$

$$= E \left[X(t) \int_{-\infty}^{\infty} h(u) X(t+\tau-u) du \right]$$

$$R_{XY}(t, t+\tau) = \int_{-\infty}^{\infty} R_{XX}(t+\tau-u-t) h(u) du$$

$$R_{XY}(t, t+\tau) = \int_{-\infty}^{\infty} h(u) R_{XX}(\tau-u) du$$

$$R_{XY}(t, t+\tau) = R_{XX}(\tau) * h(\tau)$$

$$R_{yx}(\tau) = R_{xx}(\tau) * h(-\tau)$$

ecf depends only on τ .

PDS of $Y(t)$.

$$S_{yy}(\omega) = S_{xx}(\omega) |H(\omega)|^2$$

$$S_{yy}(\omega) = F[R_{yy}(\tau)]$$

$$= \int_{-\infty}^{\infty} R_{yy}(\tau) e^{-j\omega\tau} d\tau$$

$$R_{yy}(\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{xx}(\tau + \tau_1 - \tau_2) h(\tau_1) h(\tau_2) d\tau_1 d\tau_2$$

$$S_{yy}(\omega) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} R_{xx}(\tau + \tau_1 - \tau_2) h(\tau_1) h(\tau_2) e^{-j\omega\tau} d\tau_1 d\tau_2 d\tau$$

$$\tau + \tau_1 - \tau_2 = t \quad d\tau = dt$$

$$S_{yy}(\omega) = \int_{-\infty}^{\infty} h(\tau_1) d\tau_1 \int_{-\infty}^{\infty} h(\tau_2) d\tau_2 \int_{-\infty}^{\infty} R_{xx}(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} h(z_1) e^{j\omega z_1} dz_1 \int_{-\infty}^{\infty} h(z_2) e^{-j\omega z_2} dz_2 \int_{-\infty}^{\infty} R_{xx}(t) e^{-j\omega t} dt$$

$$S_{yy}(\omega) = H^*(\omega) H(\omega) S_{xx}(\omega)$$

$$S_{yy}(\omega) = S_{xx}(\omega) |H(\omega)|^2$$

Gross Power DS of I/P & O/P

$$R_{xy}(z) = R_{xx}(z) * h(z)$$

$$F[R_{xy}(z)] = F[R_{xx}(z) * h(z)]$$

$$S_{xy}(\omega) = S_{xx}(\omega) H(\omega)$$

$$R_{yx}(z) = R_{xx}(z) * h(-z)$$

$$S_{yx}(\omega) = S_{xx}(\omega) H^*(\omega)$$

$$R_{yy}(z) = R_{xy}(z) * h(-z)$$

$$S_{yy}(\omega) = S_{xy}(\omega) \cdot H^*(\omega)$$

$$\begin{aligned} S_{yy}(\omega) &= S_{xx}(\omega) H(\omega) H^*(\omega) \\ &= S_{xx}(\omega) |H(\omega)|^2 \end{aligned}$$

$$S_{yy}(\omega) = S_{yx}(\omega) H(\omega)$$

Noise : undesired or unwanted electrical signal

White Noise / White Gaussian Noise

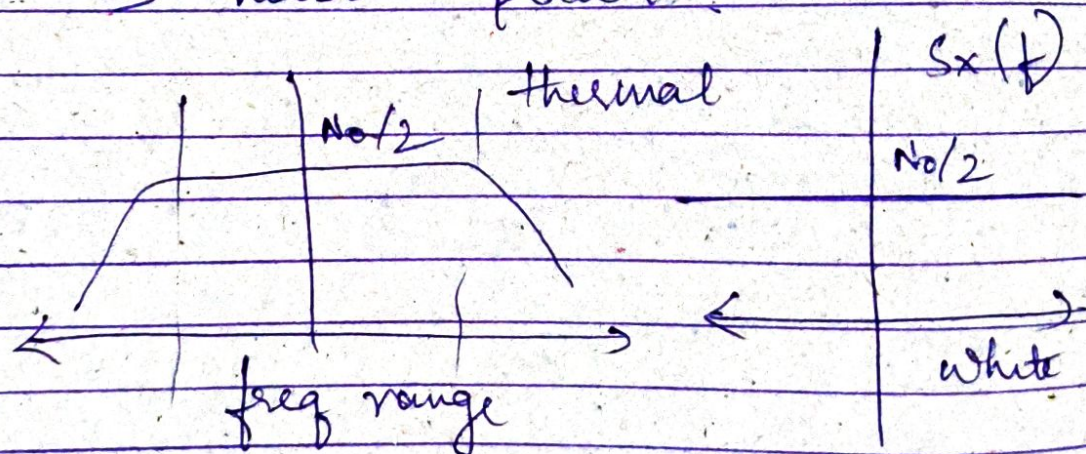
It is a thermal noise whose freq comp extend from $0 \rightarrow \infty$

WSS thermal noise process $N(t)$.

It is WN if PSD of $N(t)$ is a const at all freq

$$S_N(\omega) = \frac{N_0}{2} \text{ for all } f$$

$N_0 \Rightarrow$ noise power



It is assumed as gaussian white noise with $\mu_x = 0$ & flat PSD

$$S_{xx}(\omega) = \frac{\text{Noise Power}}{2 \times BW} = \frac{KT B}{2B} = \frac{KT}{2}$$

$$N_0 = kT \text{ watts}$$

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$$S_x(\omega) = \frac{N_0}{2} \quad \text{for} \quad -\infty < \omega < \infty$$

$$R_x(z) = F^{-1}[S_x(\omega)] = \frac{N_0}{2} \delta(z)$$

Band limited white noise

filtered white noise, finite over a band of freq

$$H(\omega) = \begin{cases} 1 & -\omega_0 \leq \omega \leq \omega_0 \\ 0 & \text{o/w} \end{cases}$$

Band Pass $H(\omega) = \begin{cases} 1 & \omega_0 - \frac{W}{2} \leq \omega \leq \omega_0 + \frac{W}{2} \\ 0 & \text{o/w} \end{cases}$

$$h(t) = 5t e^{-2t} \quad E[x(t)] = 3$$

$$E[y(t)] = ? \quad F[t e^{-at} u(t)] = \frac{1}{(a+j\omega)^2}$$

$$H(\omega) = \frac{5}{(2+j\omega)^2} \quad H(0) = \frac{5}{4}$$

$$E[y(t)] = H(0) E[x(t)] = \frac{15}{4}$$